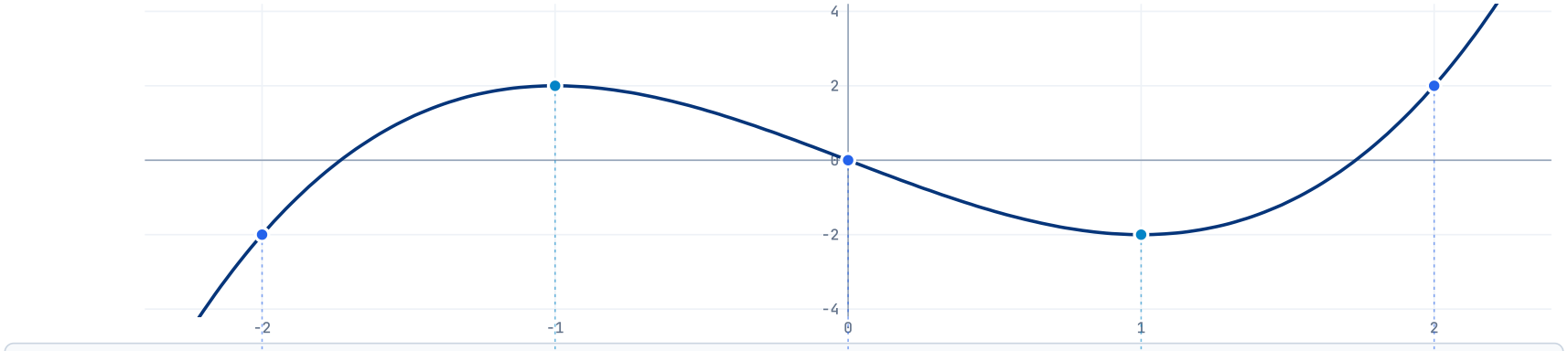
ONE FUNCTION, SIGN OF THE DERIVATIVE

Where x³ − 3x rises and falls

$f'(x) = 3x^2 - 3$

2 CRITICAL POINTS · 3 INTERVALS



KEY POINTS

1 F' = 0 AT

$x = -1, x = 1$

2 TEST VALUES

$x = -2, 0, 2$

GLOBAL

TEST RESULTS

$f'(-2) = 9,\ f'(0) = -3,\ f'(2) = 9$

INCREASING ON

$(-\infty, -1)$ and $(1, \infty)$

DECREASING ON

$(-1, 1)$